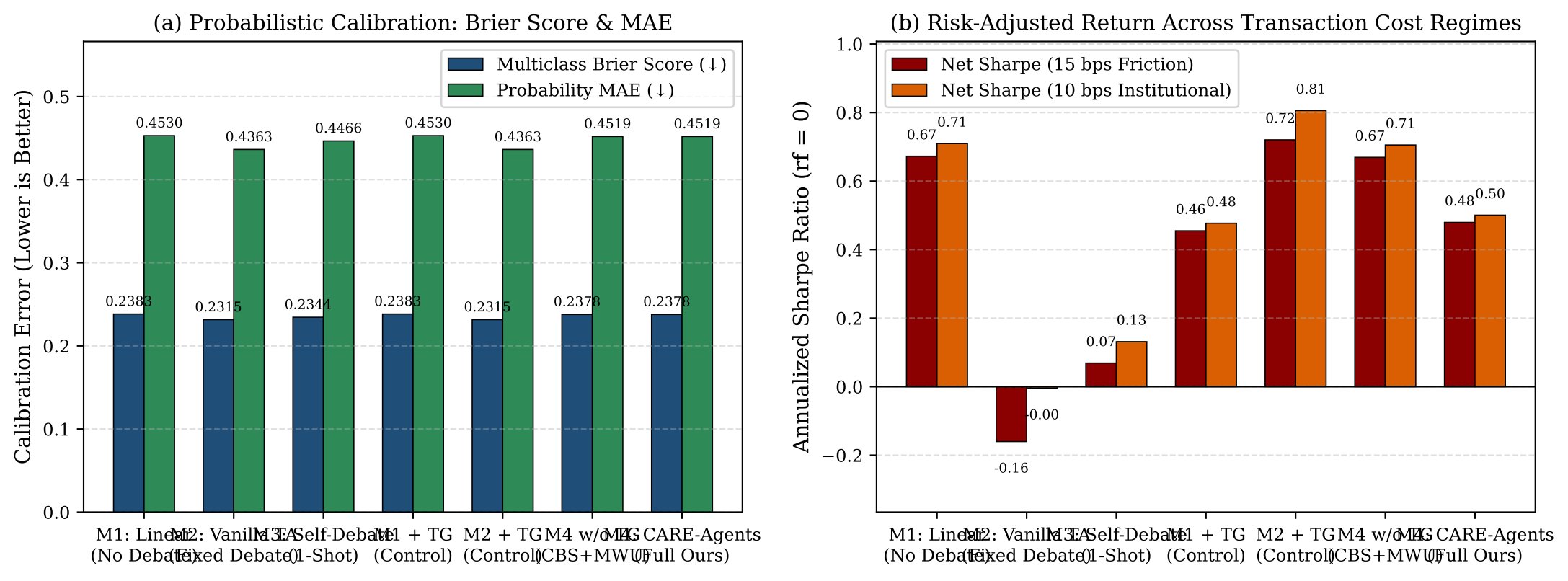
(a) Probabilistic Calibration: Brier Score & MAE
Multiclass Brier Score (↓)
Probability MAE (↓)
Calibration Error (Lower is Better)
0.0
0.1
0.2
0.3
0.4
0.5
0.2383
0.4530
0.2315
0.4363
0.2344
0.4466
0.2383
0.4530
0.2315
0.4363
0.2378
0.4519
0.2378
0.4519
M1: Lineal (No Debate)
M2: Vanilla MAS (Fixed Debate)
M3: EA Self-Debate (1-Shot)
M1 + TG (Control)
M2 + TG (Control)
M4 w/o MG (CBS+MWU)
M4: CARE-Agents (Full Ours)
(b) Risk-Adjusted Return Across Transaction Cost Regimes
Net Sharpe (15 bps Friction)
Net Sharpe (10 bps Institutional)
Annualized Sharpe Ratio (rf = 0)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
0.67
0.71
-0.16
-0.00
0.07
0.13
0.46
0.48
0.72
0.81
0.67
0.71
0.48
0.50
M1: Lineal (No Debate)
M2: Vanilla MAS (Fixed Debate)
M3: EA Self-Debate (1-Shot)
M1 + TG (Control)
M2 + TG (Control)
M4 w/o MG (CBS+MWU)
M4: CARE-Agents (Full Ours)